

We start with the inclusion $\supseteq$ and hence it is enough to show for $\mathbf{n} \leq \mathbf{m}$ aperiodic that $\mathcal{S}_{gen}(\mathbf{n}) \subseteq \mathcal{W}(\rho \times \mathcal{S}_{gen}(\mathbf{m}'))$. We first note that since $\mathbf{m}'$ and $\mathbf{m}$ are not aperiodic, we decompose $\mathbf{m} = \mathbf{m}_p + \mathbf{m}_{ap}$, where $\mathbf{m}_p = [a, b]_\rho + \dots + [a + o(\rho) - 1, b + o(\rho) - 1]_\rho$ and $[a, b]_\rho$ is the minimal $[a, b]_\rho$ found in $\mathbf{m}$ with $[a, b]_\rho + \dots + [a + o(\rho) - 1, b + o(\rho) - 1]_\rho$ contained in $\mathbf{m}$. Note that $\mathbf{m} \neq \mathbf{m}_p$, since otherwise there exists no $\mathbf{m}'$ such that $\rho * \mathbf{m}' = \mathbf{m}$. Let $\mathbf{n} \leq \mathbf{m}$ be an aperiodic multisegment. There exists $\mathbf{n}_p < \mathbf{m}_p$ obtained via one elementary operation $[c, d]_\rho + [c + 1, d + 1]_\rho \mapsto [c, d + 1]_\rho + [c + 1, d]_\rho$ such that $\mathbf{m} \leq \mathbf{n}_p + \mathbf{m}_{ap}$ by Lemma 2.2.3. Let $[1, b']_\rho$ be the segment in $\mathbf{m}'$ such that $\mathbf{m} = \mathbf{m}' - [1, b']_\rho + [0, b]_\rho$. Let us first assume that either $c \not\equiv 1 \pmod{o(\rho)}$ or $c \equiv 1 \pmod{o(\rho)}$ and $l([1, b']_\rho) \geq l([c, d + 1]_\rho)$. Then it is easy to see that there exists $\mathbf{n}'$ such that $\rho * \mathbf{n}' = \mathbf{n}_p + \mathbf{m}_p$ and $\mathbf{n}' \leq \mathbf{m}'$. In this case we have by induction

$$\mathcal{S}_{gen}(\mathbf{n}) \subseteq \mathcal{S}_{gen}(\rho * \mathbf{n}') = \mathcal{W}(\rho \times \mathcal{S}_{gen}(\mathbf{n}')) \subseteq \mathcal{W}(\rho \times \mathcal{S}_{gen}(\mathbf{m}')).$$

If $c \equiv 1 \pmod{o(\rho)}$ and $l([1, b']_\rho) \leq l([c, d + 1]_\rho)$ we obtain that $[1, b'] = [c, d]_\rho$. In this case we differentiate again $o(\rho) > 2$ and $o(\rho) = 2$. In the former, we let $\mathbf{m}'' = \mathbf{m} - [c, d + 1]_\rho - [c - 1, d]_\rho + [c + 1, d + 1]_\rho - [c, d]_\rho$ and note that $\rho|-|\rho * \mathbf{n}'' = \mathbf{m} \rho * \rho|-|\mathbf{n}'' = \mathbf{n}$. From here the claim follows as in the aperiodic case. If $o(\rho) = 2$ we let $\mathbf{m}'' = \mathbf{m} - [c, d + 1]_\rho - [c - 1, d]_\rho + [c + 2, d + 1]_\rho - [c, d]_\rho$ and note that $\rho|-|\rho * \rho * \mathbf{n}'' = \mathbf{m} \rho * \rho|-|\rho * \mathbf{n}'' = \mathbf{n}$ and we again finish the argument as in the aperiodic case.

It remains to show that $\mathcal{W}(\rho \times \mathcal{S}_{gen}(\mathbf{m}')) \subseteq \mathcal{S}_{gen}(\mathbf{m})$. Since $\mathbf{m}'$ cannot be aperiodic, it is enough to show for aperiodic $\mathbf{n} \leq \mathbf{m}'$, $\rho * \mathbf{m}' \leq \mathbf{m}$, which follows from Lemma 2.4.1.

Claim 4: $\mathbf{m}' * \rho = \mathbf{m} \Rightarrow \mathcal{W}(\rho \times \mathcal{S}_{gen}(\mathbf{m}')) = \mathcal{S}_{gen}(\mathbf{m})$

This can be proven as Claim 1 and Claim 3. □

Finally for $\mathbf{m} \in \text{Mult}_{R, \square}$, let $\mathbf{m} = \mathbf{m}_1 + \dots + \mathbf{m}_k \in \text{Mult}_R$ be a multisegment with $\mathbf{m}_i \in \text{Mult}_{\mathbb{F}_\ell}(\rho_i)$, $\rho_i \in \mathfrak{C}^\square$ and in pairwise different cuspidal lines. Then we set

$$\mathcal{S}_{gen}(\mathbf{m}) := \mathcal{W}(\mathcal{S}(\mathbf{m}_1) \times \dots \times \mathcal{S}(\mathbf{m}_k))$$

and if $\mathbf{m}' = \mathbf{m}'_1 + \dots + \mathbf{m}'_k \in \text{Mult}_{R, \square}$, $\mathbf{m}_i \in \text{Mult}_{\mathbb{F}_\ell}(\rho_i)$ is a second product, we define we

$$\mathbf{m} * \mathbf{m}' := \mathbf{m}_1 * \mathbf{m}'_1 + \dots + \mathbf{m}_k * \mathbf{m}'_k,$$

extending the product $*$ to $*\text{Mult}_{R, \square} \times \text{Mult}_{R, \square} \rightarrow \text{Mult}_{R, \square}$.

Lemma 3.4.2. *Let $\mathbf{m} = \mathbf{m}_1 + \dots + \mathbf{m}_k \in \text{Mult}_{R, \square}$ as above. Then*

$$\mathcal{W}(\mathcal{S}_{gen}(\mathbf{m})) \cong \mathcal{S}_{gen}(\mathbf{m}_1) \times \dots \times \mathcal{S}_{gen}(\mathbf{m}_k)$$

and hence for $\mathbf{m}' \in \text{Mult}_{R, \square}$ a second multisegment

$$\mathcal{W}(\mathcal{S}_{gen}(\mathbf{m}) \times \mathcal{S}_{gen}(\mathbf{m}')) = \mathcal{S}_{gen}(\mathbf{m} * \mathbf{m}').$$

Moreover, if $\mathbf{n} \leq \mathbf{m}$ then $\mathcal{S}_{gen}(\mathbf{n}) \subseteq \mathcal{S}_{gen}(\mathbf{m})$.

Proof. Note that there exists a non-zero map

$$\mathcal{S}_{gen}(\mathbf{m}_1) \times \dots \times \mathcal{S}_{gen}(\mathbf{m}_k) \rightarrow \mathcal{W}(\mathcal{S}_{gen}(\mathbf{m})).$$

Let π be the unique irreducible subrepresentation of the latter representation. It is then enough to show that π is the unique subrepresentation of the former representation. To see this note that by Theorem 2.2.1 and Lemma 2.2.1, the representation $\pi_1 \times \dots \times \pi_k$ is irreducible, where π_i is the unique irreducible subrepresentation of $\mathcal{S}_{gen}(\mathbf{m}_i)$. Since $\pi_1 \times \dots \times \pi_k$ is generic, it has to be isomorphic to π . Thus π is a subrepresentation of

$$\mathcal{S}_{gen}(\mathbf{m}_1) \times \dots \times \mathcal{S}_{gen}(\mathbf{m}_k).$$

□